Document Control

Title

Lidocaine Infusion Therapy for Chronic Neuropathic Pain Standard Operating Procedure

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Consulted with the following stakeholders: • Pain Consultants • Pain Clinical Nurse Specialists • Lead Surgical Pharmacist	**Contact responsible for implementation and monitoring compliance:** Pain Team Clinical Lead **Education/ training will be provided by:** Pain Team

Approval and Review Process

- Pain Team Clinical Lead
- Anaesthetics governance
- Divisional governance

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1. Background

Neuropathic pain is a disorder that can be accompanied by severe pain that is both persistent and resistant to conventional therapies. Lidocaine is a local anaesthetic which blocks voltage gated sodium channels within neurons to block transmission of pain. It is commonly used as a local anaesthetic injection in the skin or associated tissues for surgery as well as in nerve blocks for anaesthesia.

There is some research which demonstrates the efficacy of Lidocaine, administered as an intra-venous infusion, for both acute and persistent neuropathic pain. Patients in whom persistent neuropathic pain has been resistant to other forms of treatment may gain benefit from Lidocaine infusions.

The infusion regimens differ between contexts and also between institutions. This SOP specifically applies to Lidocaine infusions performed by a pain practitioner from the pain team as a day case procedure on the authority of a Pain Consultant.

2. Purpose

2.1. The Standard Operating Procedure (SOP) has been written to:

- Ensure that standard infusion doses are prescribed and administered in all patients with persistent neuropathic pain requiring Lidocaine intravenous infusion.
- Ensure that intravenous Lidocaine infusions are administered in a safe manner, correctly prescribed and administered, using the correct equipment.
- Ensure that appropriate monitoring of patients using intravenous Lidocaine infusions occurs and that actions in event of adverse reactions are described and equipment available.

3. Scope

3.1. This Standard Operating Procedure (SOP) relates to the following staff groups who may be involved in the assessment and delivery of pain relief:

- Pain Consultants and pain practitioners who could include: Clinical Nurse Specialist; Operating Department Practitioner or other registered health professionals within the pain team.
- <u>Initial assessment of patients for Intravenous Lidocaine Infusion</u>:
- Initially a pain consultant or pain practitioner should assess the patient and consent for treatment with an intravenous Lidocaine infusion as a day case procedure to be booked at a separate appointment.
- On occasion, a pain consultant can assess relevant in-patients at ward level, on a case by case basis with pain practitioner involvement.

4. Location

4.1. This Standard Operating Procedure regarding intravenous Lidocaine infusions for persistent neuropathic pain can be implemented where competent members of staff are available to undertake this role where the required levels of patient monitoring and standard resuscitation facilities are available.

4.2. Staff undertaking this procedure must be able to demonstrate continued competence. As the primary dangers are those of local anaesthetic toxicity, only pain practitioners trained to recognise and manage local anaesthetic toxicity will administer the infusion. This also applies at ward level.

5. Equipment

- Blue and pink safety cannulas
- 2% chlorhexidine gluconate and 70% alcohol swab
- Disposable tourniquet
- IV 3000 cannula dressing
- IV cannula extension device
- 1 x 50 ml luerlok syringe
- 1 x red or purple filter needle for drawing up
- 0.9% sodium chloride solution
- 1% and 2% 5 ml Lidocaine vials
- 1 x infusion giving set including anti syphon
- Equipment to monitor: Blood pressure, pulse, saturations, ECG and pain score every 10-15 minutes
- Intravenous Lidocaine must be administered via a programmable infusion device, at a fixed rate, according to the protocol. It must be delivered via a sole use dedicated cannula with an anti-syphon line.
- Two bags of 500 ml Intra-Lipid 20% must be immediately available, along with the AAGBI guideline for treatment of local anaesthetic toxicity. (See Paragraph 8).

6. Procedure

- **Patient selection criteria for Intravenous Lidocaine: Must be made by a pain consultant or pain practitioner at either clinic or ward level for patients :**
- With persistent neuropathic pain that has failed to benefit from opioid and anti-neuropathic medication.

- With persistent neuropathic pain including, but not exclusively, the following conditions:

  Multiple Sclerosis

  Trigeminal neuralgia

  Chemo-induced neuropathy

  Complex Regional Pain Syndrome (CRPS)

  It is not for use in generalised widespread pains such as Fibromyalgia or Arthritis.

  Caution should be applied in patients with epilepsy or cardiac disease with risk of arrhythmia.

  It is avoided in pregnant women.

- Written consent is required: this must be obtained by a pain consultant or pain practitioner and filed in the patient's medical notes.
- Lidocaine infusion for persistent neuropathic pain is an 'off label' use of the drug and is therefore not covered by the manufacturer's license.
- Pain consultants or pain practitioners, who are non-medical prescribers may prescribe intravenous Lidocaine for the management of persistent neuropathic pain and therefore assume responsibility for its use.
- Patients must be involved in the decision process and be made aware of the risk of local anaesthetic toxicity. These are usually in the form of feelings of light headedness, metallic taste, tinnitus and peri-oral tingling. Lidocaine has a good safety profile with slow progression of identifiable symptoms and signs prior to reaching toxic levels. Patients must be warned of the risk of toxicity which can in severe cases result in seizures, loss of consciousness and in extreme cases cardiac arrest.
- Patient leaflet is to be given to the patient at the time of consenting.

- The patient must be weighed prior to writing the prescription.

7. Prescribing/Administration Instructions:

- The infusion must be prescribed on an approved in-patient prescription chart.

- The dose is 5 mg/kg made up to 50 ml with 0.9% sodium chloride solution. (Maximum dose 500mg in total).

- Infusions are started at 25ml per hour via a syringe driver infusion pump. If patients have tolerated 25ml/hour on previous infusions the rate may be cautiously increased up to 50ml per hour

- If the patient develops peri-oral tingling, metal taste or feels light headed the infusion is reduce the infusion rate by 50%. If symptoms persist after 15 minutes, a further reduction to the infusion rate of 30% is indicated

- Minimum infusion running time is 1 hour. Maximum duration of infusion running time is 2 hours.

- Total amount given and running time of infusion is to be documented in the patient notes.

- In the unlikely event of seizures, cardiac arrest or if the patient feels very unwell, the infusion must be stopped immediately. The anaesthetic consultant on call or pain consultant should be notified. Supportive care must commence immediately as detailed below, with standard life support in cardiac arrest.

- Blood pressure, pulse, saturations and respirations must be monitored every 10 minutes for the first half an hour and then subsequently at 15 minute intervals for the duration of the infusion. In addition, patients need to be assessed for symptoms of local anaesthetic toxicity such as peri-oral tingling, metallic taste, tinnitus, feeling light headed and ECG changes every 10 minutes for the first half an hour and then subsequently at 15 minute intervals for the duration of the infusion. These are to be recorded on the approved Lidocaine Infusion record sheet (see **appendix 1)**.

8. Management of adverse effects

Metal taste / peri-oral tingling / feeling light headed

- Reduce infusion rate by 50%
- Check drug prescription (specific to weight) and pump infusion rate
- Contact Pain consultant
- If symptoms persist after 15 minutes reduce by a further 30%
- If symptoms worsen STOP the infusion

Seizures / loss of consciousness

- Stop Lidocaine infusion
- Initiate supportive care and manage the patient's airway
- If unable to contact pain consultant, fast bleep on call anaesthetic consultant

Cardiac Arrest

- Immediately initiate life support with cardio-pulmonary resuscitation (CPR)
- Stop Lidocaine infusion
- Call cardiac arrest team
- Administer intra-lipid 20% 1.5 ml / kg and initiate infusion at 15 ml / kg / hour as per AAGBI guidelines on local anaesthetic toxicity
- Contact Pain consultant

9. Follow up Arrangements:

- Patients must remain in the area for observation for 30 minutes after ceasing the infusion.

- Discharge summary to be completed within 24 hours of discharge from the day case ward.

- Patient must be accompanied home by a responsible adult who must remain with them for at least 12 hours. The patient may not drive for 24 hours.

- Outcome measures will be recorded in the patients notes using the approved Lidocaine follow-up outcome measure (**see appendix 2**). If the patient has gained at least a 30% reduction in pain related symptoms the patient will be re-booked for further infusions. If there has been less than a 30% reduction in pain related symptoms the patient will be referred back to the original pain consultant for review of management plan.

10. References

- Cochrane review: Systemic administration of local anaesthetic agents to relieve neuropathic pain, 2005: http://onlinelibrary.wiley.com/doi/10.1002/14651858.CD003345.pub2/full

- Association of Anaesthetists of Great Britain and Ireland (AAGBI) guidelines on management local anaesthetic toxicity: https://www.aagbi.org/sites/default/files/la_toxicity_2010_0.pdf

- The Efficacy of Systemic Lidocaine in the Management of Chronic Pain: A Literature Review. Anesth Pain Med. 2017 June; 7(3):e44732. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5561441/
- Intravenous Lidocaine in the Management of Various Chronic Pain States: A Review of 211 Cases. Regional Anaesthesia and Pain Medicine. 1985. https://rapm.bmj.com/content/10/1/1

11. Associated Documentation

- NDHT Injectable Medicines Policy: http://www.northdevonhealth.nhs.uk/wp-content/uploads/2016/08/Injectable-Medicines-Policy-v1.4.pdf

- NDHT Resuscitation Policy: http://www.northdevonhealth.nhs.uk/wp-content/uploads/2014/02/Resuscitation-Policy-V5.0-28Oct14.pdf

- NDHT Consent Policy: http://www.northdevonhealth.nhs.uk//wp-content/uploads/2012/04/Consent-Policy-v3.3-28Aug14.pdf

- NDHT Lidocaine Infusion Patient Information Leaflet: https://www.northdevonhealth.nhs.uk/wp-content/uploads/2019/09/Lidocaine-infusion.pdf

Appendix 1

NHS
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Affix Patient label

Intravenous Lidocaine Documentation Record

Date:

Treatment Number:

Diagnosis:

Written consent: Yes/No

Patient leaflet: Yes/No

Cons Name:

Bleep No:

Analgesic taken pre-Lidocaine infusion

Pain Score pre infusion (using outcome form):

Allergies:

Setting Up Infusions

Cannula: (Insert Sticker)

Size/Site:

Inserted by:

Lidocaine

Patients weight :…….(kg)

Dose prescribed: …… (mg)

(See prescription chart)

Rate: …… (mls/hr)

Lidocaine Dose/Infusion Rate:

5mg/kg made up to 50mls with Normal Saline.

Any recorded problems with previous treatments:

Intralipid Expiry date:……

ECG Monitoring in situ Yes/No

Observations during Intravenous Lidocaine Infusion

Time	N.I.B.P	H.R	RR/Sa02	Side effect Neuro	Side Effect Cardio	SIG.
Start of infusion:						
+10 min for 30 mins						
+15 mins for duration.						

Time Infusion Stopped…… Total amount given (mls)……(mg)…….

Cannula Removed Y/N Post Infusion pain score……..

Refreshment's given Y/N

Time Discharged…… Accompanied by…….

Discharge Summary Completed? Y/N.

Signature..... Print..... Title.....

Appendix 2

Lidocaine Follow-up Outcome Measure

Pain Outcome Measure				
Do you feel confident that you can cope with your pain?	Every day = 0	Most days = 1	Some days = 2	Never = 3
Do you find it easy to relax and/or distract yourself from your pain?	Every day = 0	Most days = 1	Some days = 2	Never = 3
Do you feel confident that you can be active and do things you enjoy whilst living with pain?	Every day = 0	Most days = 1	Some days = 2	Never = 3
How would you rate your pain over the last few weeks?	No pain = 0	Mild pain = 1	Moderate pain = 2	Severe pain = 3
At its best				
At its worst				
How much does your pain interfere with your …	No interference = 0	Mild interference = 1	Moderate interference = 2	Severe interference = 3
Mood				
Relationships (friends & family)				
Work (paid & voluntary)/Hobbies				
Sleep/rest				
General activity				
Total score				/30

If the patient has gained at least a 30% reduction in pain related symptoms the patient will be re-booked for further infusions.

If there has been less than a 30% reduction in pain related symptoms the patient will be referred back to the original pain consultant for review of management plan.